

COURSE CONTENTS

Category	Title	Code	Credits-4C			Theory Papers
Basic Sciences BS-5	Mathematics-III	BE 301	L	T	P	Max Marks-100
			3	1	0	Min Marks-35 Duration-3 Hrs

Unit-I Functions of Complex Variables: Analytic functions, Harmonic Conjugate, Cauchy - Riemann Equations, Line integral, Cauchy's theorem, Cauchy's Integral formula, Singular points, Poles and Residues, Residue theorem, Evaluation of Real Integral, Bilinear Transformation.

Unit-II Numerical Analysis: Difference operators, Errors and Approximations, Interpolation, Inverse interpolation, Numerical differentiation, Numerical Integration by using Simpson's method, Weddel's rule and Gauss legendre open quadrate formula.

Unit-III Solutions of algebraic and transcendental equations(Regular False, Newton-Raphson, Iterative, Graffe's root squaring methods), Solutions of simultaneous algebraic equations, Solutions of ordinary differential equations (Tailor's Series, Picard's Method, Modified Euler's method, Runge Kutta Method, Predictor-Corrector Method), Solution of Partial differential equation.

Unit-IV Linear Algebra: Simultaneous linear equations, Vector space and subspaces, Bases and dimension, Linear transformations and their matrix representations, Change of basis

Unit-V Diagonalizable operators, Quadratic forms and their digitalization, Canonical form, Bilinear, Hermitian forms, Definite and semi definite forms.

References:

1. Chandrasekharaiah DS; Engineering Maths Part II & III; Prism Books Pvt.
2. Kreyszig E; Advanced Engineering Mathematics; Wiley Eastern Limited.
3. Stanton RJ; Numerical Analysis
4. Grewal BS; Higher Engineering Mathematics; Khanna Publisher.
5. Jaggi, Mathur; Engineering Mathematics; Khanna Publisher.

COURSE CONTENTS

Category	Title	Code	Credits-4C			Theory Papers
Humanities and Science HS 2	Energy Environment Ethics and Society	BE 302	L	T	P	Max Mark-100 Min Mark-35 Duration-3Hrs
			3	1	0	

Unit-I Energy: linkage with development, world energy scenario, fossil fuel resource-estimates and duration, India's energy scenario; Finite/ depleting energy resources, coal, oil, gas, nuclear fission, promises and present status of nuclear fusion energy; Renewable energy, solar, hydro, wind, biomass, ocean, tidal, wave and geothermal.

Unit -II Society and environment: exponential growth in population, environmentally optimum sustainable population, pros and cons of reducing birth-rates, solar and natural capital, perpetual, renewable and exhaustible resources, free access resources and the tragedy of commons, environment problems and impact of P.A.T (Population, Affluence and Technology), environmentally beneficial and harmful technologies, interaction between environment life support systems and socio-culture system.

Unit-III Ecosystem definition, concepts, structure, realm of ecology, lithosphere, hydrosphere, biosphere, atmosphere-troposphere-stratosphere; Nonrandom high quality solar energy flow/ balance to earth, greenhouse effect, matter and nutrient recycling in ecosystems; nitrogen, oxygen, carbon and water cycles, food producers, consumers and decomposers, food chains; biodiversity, threat and conservation of biodiversity.

Unit-IV Air pollution-primary, secondary, chemical and photochemical reactions, effects of CO, NO, CH and particulates, acid rain, global warming and Ozone depletion; monitoring and control of pollutants; noise pollution-sources and control measures; thermal-, heavy metals- and nuclear pollutions; industrial pollution from paper, pharmacy, distillery, tannery, fertilizer, food processing and small scale industries. Environment impact assessment policies and auditing, conflicting worldviews and environmentally sustainable economic growth, introduction to Design For Environment (DFE), product lifecycle assessment for environment and ISO 14000; triple bottom-line of economic, environment and social performance;

Unit-V Ethics and moral values, ethical situations, objectives of ethics and its study, role morality and conflicts, values, policy and Organization Culture; Non- quasi- and hard-professionals; preventive, personal, common and professional ethics, different ethical value criteria with discussion on the case of priority for improvement of urban (high traffic) or rural (low traffic) intersections causing equal number of fatalities, codes of ethics and their limitation; Institute of engineers code for corporate member and ACM professional-code.

References:

1. Miller G. T Jr: Living in the environment; Cengage Publisher.
2. Harris CE, Prichard MS, Rabins MJ: Engineering Ethics, Cengage Pub.
3. Govindrajan, Natrajan, Santikumar: Engineering Ethics, PHI pub.
4. Rana SVS: Essentials of ecology and environment, PHI Pub.
5. Gerard Kiely: Environmental Engineering, TMH
6. Khan BH: Non Conventional energy resources, TMH Pub.
7. Raylond G.W.: Ethics in Information Technology, Cengage.

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
DC-1	Electronic Instrumentation	EC--303	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	2	

Unit-I Measurement and Error, Accuracy and Precision, Sensitivity, Linearity, Resolution, Hysteresis, Loading Effect. Measurements of Current, Voltage, Power and Impedance: DC and AC Ammeter, DC Voltmeter-Chopper type and solid-state, AC voltmeter using Rectifier, Average, RMS, Peak Responding voltmeters, Multi-meter, Power meter, Bolometer and Calorimeter.

Unit-II Cathode Ray Oscilloscope (CRO): Different parts of CRO, Block diagram, Electrostatic focusing, Electrostatic deflection, Post deflection acceleration, Screen for CRTs, Graticules, Vertical and Horizontal deflection system, Time base circuit, Oscilloscope Probes, Applications of CRO, Special purpose CROs-Multi input, Dual trace, Dual beam, Sampling, Storage (Analog and Digital) Oscilloscope.

Unit-III AC Bridges: Maxwell's bridge (Inductance and Inductance-Capacitance), Hay's bridge, Schering bridge (High voltage and Relative permittivity), Wein bridge, Wagner earth detector, Impedance measurement by Q-meter. **Non-Electrical Quantities (Transducer)**: Classification of Transducers, Strain gauge, Displacement Transducer-Linear Variable Differential Transformer (LVDT) and Rotary Variable Differential Transformer (RVDT), Temperature Transducer- Resistance Temperature Detector (RTD), Thermistor, Thermocouple, Piezo-electric transducer, Optical Transducer- Photo emissive, Photo conductive, Photo voltaic, Photo-diode, Photo Transistor, Nuclear Radiation Detector.

Unit-IV Wave Analyzer (Frequency selective and Heterodyne), Harmonic Distortion Analyzer, Spectrum Analyzer, Signal and Function Generators, Sweep Frequency Generator, Pulse and Square Wave Generator, Beat Frequency Oscillator.

Unit-V Digital Measurement and Instruments: Advantages of Digital Instrument over Analog Instrument, Digital-to-analog conversion (DAC) - Variable resistive type, R-2R ladder Type, Binary ladder, Weighted converter using Op-amp and transistor, Practical DAC. Analog-to-digital Conversion (ADC) -Ramp Technique, Dual Slope Integrating Type, Integrating Type (voltage to frequency), Successive Approximations, digital voltmeters and multi-meters, Resolution and sensitivity of digital meter, PLC structure, principle of operation, response time and application.

References:

1. H. S. Kalsi: Electronics Instrumentation, TMH.
2. K. Sawhney: Instrumentation and Measurements, Dhanpat Rai and Co.
3. Helfric and Cooper: Modern Electronic Instrumentation and Measurement Techniques; Pearson.

List of Experiments:

All experiments (wherever applicable) should be performed through the following steps.

Step 1: Circuit should be designed/drafted on paper.

Step 2: The designed/drafted circuit should be simulated using Simulation Software (TINA-PRO/ PSPICE/ LABVIEW/ CIRCUIT MAKER).

Step 3: The designed/drafted circuit should be tested on the bread board and compare the results with the simulated results.

Step 4: The bread board circuit should be fabricated on PCB by one batch using PCB machine.

1. Study of CRO and Function Generator.
2. Displacement measurement by LVDT.
3. Force measurement by strain gauge.
4. Measurement of Capacitor, Self-induction using Q-meter.
5. Temperature measurement by thermistor, RTD and thermocouple.
6. Optical Transducer- Photo conductive, Photo voltaic, Photo-diode, Photo-Transistor
7. Design of digital to analog converter.
8. PLC operation and applications (for example: relay, timer, level, traffic light etc.)

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
Department DC-2	Electronic Devices	EC/BM--304	L	T	P	Max. Marks-100
			3	1	2	Min. Marks: 35 Duration: 3 hrs.

Unit-I Semiconductor intrinsic and extrinsic, p-type and n-type, energy band diagrams, majority and minority carrier, charge density in semiconductor, generation and recombination of charges, process of diffusion, diffusion and drift currents, Hall effects and its applications. p-n junction, depletion layer, potential barrier, electric field, forward and reverse biased junction, current components in p-n diode, current equation, V-I characteristics, cut in voltages of Si and Ge diode, transition and diffusion capacitance, power dissipation, p-n junction diode as rectifier, clipper and clamper.

Unit-II Optoelectronic and miscellaneous devices: Characteristics and application of Zener diode, Varactor diode, Schottky diode, Tunnel Diode, PIN diode, LED, photoconductor cells, photodiodes, solar cell, phototransistors, opto-couplers, thermistors, Seven segment displays.

Unit-III Bipolar junction transistor - Construction, basic operation, current components and equations, CB, CE and CC-configuration, input and output characteristics, Early effect, region of operation, active, cutoff and saturation region Ebers-Moll model, power dissipation in transistor ($P_{d_{max}}$ rating), Uni-junction Transistor (UJT) : Principle of operation, characteristics.

Unit-IV FET construction- Construction, n channel and p channel, characteristics, parameters, equivalent model and voltage gain, Enhancement and depletion MOSFET and its characteristics, analysis of FET in various configuration.

Unit-V Thyristor Family- Silicon Controlled Rectifier, V-I Characteristics, Transistor Analogy, Turn-On and Turn-Off Mechanism, Series And Parallel Combination of SCR, Protection Circuits. Introduction to Diac, Triac, Power MOSFET, IGBT and GTO.

References:

1. Boylestad and Nashelsky : Electronic Devices and Circuit Theory, Pearson Education
2. Millman and Halkias : Integrated electronics, TMH
3. Graham Bell : Electronic Devices and Circuits , PHI
4. Ned Mohan : Power electronics, John Wiley and Sons
5. Sendra and Smith : Microelectronics, Oxford Press.
6. Streetman : Electronic Devices, Pearson Education.
7. Donald A Neamen : Electronic Circuits Analysis and Design, TMH
8. Salivahanan et al : Electronic Devices and Circuits, TMH

List of Experiments (Expandable):

All experiments (wherever applicable) should be performed through the following steps.

Step 1: Circuit should be designed/drafted on paper.

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Step 4: The bread board circuit should be fabricated on PCB by one batch using PCB machine.

1. V-I characteristics of various Diodes (p-n, Zener, Varactor, Schottky, Tunnel, Photodiode etc)
2. Characteristics of Transistors (BJT and FET)
3. Study of Power electronic devices (Diac, Triac, SCR, Power MOSFET, IGBT etc.)

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
DID-1	Network Analysis	EC--305	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	2	

Unit-I Introduction to circuit elements R, L, C and their characteristics in terms of linearity and time dependence, KCL and KVL analysis, dual networks, analysis of magnetically coupled circuits, Dot convention, coupling co-efficient, Tuned circuits, Series and parallel resonance, voltage and current sources, controlled sources.

Unit-II Network topology, Concept of Network graph, Tree, tree branches and links, cut set and tie set schedules. Network Theorems – Thevenin, Norton, Superposition, Reciprocity, Compensation, Maximum power transfer and Millmans theorems, problems with controlled sources.

Unit-III Transient analysis: Transients in RL, RC and RLC circuits, initial conditions, time constants, networks driven by constant driving sources and their solutions.

Steady state analysis: - Concepts of phasors and vectors, impedance and admittance. Node and mesh analysis of RL, RC and RLC networks with sinusoidal and other driving sources. Resonance Circuits.

Unit-IV Frequency domain analysis – Laplace transform solution of Integral-differential equations. Transform of waveform – step, ramp, Gate and sinusoidal functions. Initial and final value theorem. Network Theorems in frequency domain. Fourier Series, Trigonometric & exponential form of fourier series, Fourier series of basic functions.

Unit-V Network function & Two port networks concept of complex frequency. Network functions of one and two ports, poles and zeros network of different kinds. Necessary conditions for driving point & transfer function.

Two port parameters– Z, Y, ABCD, hybrid parameters, their inverse and image parameters, relationship between parameters. Interconnection of two port networks, Terminated two port networks.

References:

1. M.E. Van Valkenburg: Network Analysis, PHI
2. F.F.Kuo: Network Analysis, Wiley India.
3. Mittal GK: Network Analysis, Khanna Publisher
4. Mesereau and Jackson: Circuit Analysis- A system Approach, Pearson.
5. Hayt W.H. & J.E. Kemmerly: Engineering Circuit Analysis, TMH
6. Decarlo lin: Linear circuit Analysis, Oxford
7. William D Stanley : Network Analysis with Applications, Pearson Education
8. Roy Choudhary D: Network and systems, New Age Pub
9. Sudhakar & Pillai: Circuit & Networks- Analysis and Synthesis, TMH

List of experiments (Expandable)

All experiments (wherever applicable) should be performed through the following steps.

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Step 2: The designed/drafted circuit should be simulated using Simulation Software (TINA-PRO/ PSPICE/ LABVIEW/ CIRCUIT MAKER).

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Step 4: The bread board circuit should be fabricated on PCB by one batch using PCB machine.

1. To Verify Thevenin Theorem.
2. To Verify Superposition Theorem.
3. To Verify Reciprocity Theorem.
4. To Verify Maximum Power Transfer Theorem.
5. To Verify Millman's Theorem.
6. To Perform Open Circuit Test on Two Port Network.
7. To Perform Short Circuit Test on Two Port Network.
8. To Find Frequency Response of LRC Series Circuit.
9. To Find Frequency Response of LRC parallel Circuit.

COURSE CONTENTS

Category	Title	Code	Credit-4			Practical
IT-2	CP III Java	CS/EC -- 306	L	T	P	Max. Marks-50 Min. Marks: 25 Duration: 3 hrs.
			-	-	4	

UNIT-I Basic Java Features - C++ Vs JAVA, JAVA virtual machine, Constant & Variables, Data Types, Class, Methods, Objects, Strings and Arrays, Type Casting, Operators, Precedence relations, Control Statements, Exception Handling, File and Streams, Visibility, Constructors, Operator and Methods Overloading, Static Members, Inheritance: Polymorphism, Abstract methods and Classes

UNIT-II Java Collective Frame Work - Data Structures: Introduction, Type-Wrapper Classes for Primitive Types, Dynamic Memory Allocation, Linked List, Stack, Queues, Trees, Generics: Introduction, Overloading Generic Methods, Generic Classes, Collections: Interface Collection and Class Collections, Lists, Array List and Iterator, Linked List, Vector.
Collections Algorithms: Algorithm sorts, Algorithm shuffle, Algorithms reverse, fill, copy, max and min Algorithm binary Search, Algorithms add All, Stack Class of Package java. Util, Class Priority Queue and Interface Queue, Maps, Properties Class, Un-modifiable Collections.

UNIT-III Advance Java Features - Multithreading: Thread States, Priorities and Thread Scheduling, Life Cycle of a Thread, Thread Synchronization, Creating and Executing Threads, Multithreading with GUI, Monitors and Monitor Locks. Networking: Manipulating URLs, Reading a file on a Web Server, Socket programming, Security and the Network, RMI, Networking, Accessing Databases with JDBC: Relational Database, SQL, MySQL, Oracle

UNIT-IV Advance Java Technologies - Servlets: Overview and Architecture, Setting Up the Apache Tomcat Server, Handling HTTP get Requests, Deploying a web Application, Multitier Applications, Using JDBC from a Servlet, Java Server Pages (JSP): Overview, First JSP Example, Implicit Objects, Scripting, Standard Actions, Directives, Multimedia: Applets and Application: Loading, Displaying and Scaling Images, Animating a Series of Images, Loading and playing Audio clips

UNIT-V Advance Web/Internet Programming (Overview): J2ME, J2EE, EJB, XML.

References:

1. Deitel & Deitel: JAVA, How to Program, 6th Edition .LPE, Pearson; ISBN 81-297-1195-8
2. Herbert Schildt: The Complete Reference, Mc Graw Hill
3. E. Balaguruswamy: Programming In Java,, 2nd Edition, TMH Publications ISBN 0-07-4635425
4. Peter Norton: Peter Norton Guide To Java Programming, Techmedia; ISBN 81-87105-61-5.
5. Merlin Hughes, Maria Winslow, Conrad Hughes, Michael Shoffner: Java Network Programming Manning Publications/Prentice Hall

List of experiments (Expandable)

1. Installation of J2SDK.
2. Write a program to show Scope of Variables.
3. Write a program to show Concept of CLASS in JAVA.
4. Write a program to show Type Casting in JAVA.
5. Write a program to show How Exception Handling is in JAVA.
6. Write a Program to show Inheritance.
7. Write a program to show Polymorphism.
8. Write a program to show Access Specifiers (Public, Private, Protected) in JAVA .
9. Write a program to show use and Advantages of CONSTRUCTOR.
10. Write a program to show Interfacing between two classes.
11. Write a program to Add a Class to a Package.
12. Write a program to show Life Cycle of a Thread.
13. Write a program to demonstrate AWT.
14. Write a program to Hide a Class.
15. Write a Program to show Data Base Connectivity Using JAVA.
16. Write a Program to show "HELLO JAVA " in Explorer using Applet.
17. Write a Program to show Connectivity using JDBC.
18. Write a program to demonstrate multithreading using Java.
19. Write a program to demonstrate applet life cycle.
20. Write a program to demonstrate concept of servlet.